

Non-equilibrium strategies enabling ligand specificity by signaling receptors

Reviewed Preprint

v2 • October 6, 2025

Revised by authors

Reviewed Preprint

v1 • July 29, 2025

Andrew Goetz , Jeremy Barrios, Ralitsa Madsen, Purushottam Dixit 

Department of Biomedical Engineering, Yale University, New Haven, United States • Department of Physics, Yale University, New Haven, United States • MRC Protein Phosphorylation and Ubiquitylation Unit, University of Dundee, Dundee, United Kingdom • Systems Biology Institute, Yale University, West Haven, United States

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This study presents a **valuable** finding about how receptor-ligand binding pathways with multi-site phosphorylation can show non-monotonic responses to increasing ligand affinity and to kinase activity. The authors provide **compelling** evidence through a simple ordinary differential equation model of such signaling networks with the key new ingredient of ligand-induced receptor degradation. The work will be of interest to physicists and biologists working on signal transduction and biological information processing.

<https://doi.org/10.7554/eLife.107524.2.sa3>

Abstract

Signaling receptors often encounter multiple ligands and have been shown to respond selectively to generate appropriate, context-specific outcomes. At thermal equilibrium, ligand specificity is limited by the relative affinities of ligands for their receptors. Here, we present a non-equilibrium model in which receptors overcome thermodynamic constraints to preferentially signal from specific ligands while suppressing others. In our model, multi-site phosphorylation and active receptor degradation act in concert to regulate ligand specificity, with receptor degradation providing a previously under-appreciated layer of control. Here, ligand-bound receptors undergo sequential phosphorylation, with progression restarted by ligand unbinding or receptor turnover. High-affinity complexes are *kinetically* sorted toward degradation-prone states, while low-affinity complexes are sorted towards inactivated states, both limiting signaling. As a result, network activity is maximized for ligands with intermediate affinities. This mechanism explains paradoxical experimental observations in receptor tyrosine kinase (RTK) signaling, including non-monotonic dependence of signaling output on ligand affinity and kinase activity. Given the ubiquity of multi-site phosphorylation and ligand-induced degradation across signaling receptors, we propose that *kinetic sorting* may be a general non-equilibrium ligand-discrimination strategy used by multiple signaling receptors.

Introduction

Signaling receptors routinely encounter a wide variety of extracellular ligands and decode their identity with remarkable precision to generate context-specific responses. This selective processing of environmental cues is essential for regulating diverse biological processes, including development, immune surveillance, and tissue homeostasis (Cantley et al., 2014). Failures in ligand discrimination underlie many diseases, including diabetes and cancer (Madsen et al., 2025; Madsen and Vanhaesebroeck, 2020).

A key determinant of ligand specificity in biochemical networks is the thermodynamic stability of molecular complexes, such as ligand–receptor or substrate–enzyme pairs. At thermal equilibrium, the abundance of complexes is determined by their equilibrium binding constants. This imposes a fundamental limit on specificity: high-affinity ligands are inevitably favored over lower-affinity competitors, with complex abundances scaling in proportion to their association constants. Notably, many biochemical networks display paradoxical behaviors that cannot be explained by equilibrium affinity alone (Clark et al., 1999; Coombs et al., 2002; Freed et al., 2017; Madsen et al., 2025; Myers et al., 2023). For example, signaling receptors such as receptor tyrosine kinases (RTKs) and T cell receptors can produce stronger signaling outputs (phosphorylation levels) in response to intermediate-affinity ligands compared to low- and high-affinity ligands (Coombs et al., 2002; Lever et al., 2014; Freed et al., 2017; Madsen et al., 2025; Myers et al., 2023). Additionally, RTKs also exhibit a non-monotonic dependence between receptor activity and kinase activity (Kiyatkin et al., 2020; Kleiman et al., 2011). These observations raise a fundamental question: how do signaling receptors overcome thermodynamic constraints to achieve robust, ligand-specific responses?

A classic scheme to bypass limitations imposed by equilibrium thermodynamics is kinetic proof-reading (KPR), a mechanism first proposed by Hopfield (Hopfield, 1974) and Ninio (Ninio, 1975). KPR enhances specificity of high affinity ligands by introducing energy-consuming, irreversible steps, such as phosphorylation/dephosphorylation cycles, that amplify differences between competing ligands. KPR has been invoked in diverse systems, including DNA replication (Hopfield, 1980), mRNA surveillance (Hilleren and Parker, 1999), protein folding (Gulukota and Wolynes, 1994), and immune receptor signaling (McKeithan, 1995; Huang et al., 2019; Lever et al., 2014). Notably, while most KPR models prefer ligands with the highest affinity, it is also known that embedding KPR schemes in larger biochemical networks may allow non-monotonic dependence between ligand affinity and network activity (Lever et al., 2014; Murugan et al., 2014). However, as we will show below, these models do not capture the non-monotonic dependence between network output and kinase activity.

In our model, high-affinity ligand-receptor complexes are sorted towards degradation-prone states, while low-affinity complexes repeatedly dissociate the ligand, resulting in maximal signaling output only from intermediate-affinity ligands. Notably, this ligand specificity can be tuned by varying easily controllable cellular parameters, e.g. enzyme abundances. This non-equilibrium *kinetic sorting* mechanism explains the paradoxical non-monotonic dependence of signaling activity on ligand affinity and phosphorylation rate observed in RTKs. More broadly, given the ubiquity of the signaling motifs involved, we propose that kinetic sorting provides a general strategy for achieving ligand discrimination that is likely to be broadly used across diverse signaling networks.

Results

Classic kinetic proofreading favors high-affinity ligands

Kinetic proofreading (KPR) is the *standard model* for non-equilibrium ligand discrimination. To set the stage, we first revisited the classic KPR model originally proposed by McKeithan to explain how T cell receptors avoid activation downstream of weak ligands (McKeithan, 1995 [\[1\]](#)) (Fig. 1a [\[2\]](#); see Supplementary Materials for equations).

In this model, ligand-bound receptors undergo a series of phosphorylation steps, with the final state P_N representing the active, signaling-competent form. Importantly, ligand unbinding at any phosphorylation stage returns the receptor to the unbound state R . We parameterized the model using dimensionless quantities: the ligand dissociation rate $\delta = k_d\tau$, phosphorylation rate $\omega = k_p\tau$, and ligand concentration $u = L/K_D$, where $K_D = k_d/k_{on}$. Assuming saturating ligand ($u \rightarrow \infty$), the steady-state abundance of the active state is:

$$P_N = \frac{\omega^N}{(\omega + \delta)^N}. \quad (1)$$

As expected, increasing the phosphorylation cascade length N amplifies the preference for low-dissociation (high-affinity) ligands (Fig. 2a [\[3\]](#)), reflecting the classical KPR outcome.

Modified KPR schemes do not explain paradoxical RTK behavior

Before introducing our model, we briefly review two previously proposed extensions of receptor-level KPR that exhibit non monotonic ligand discrimination: the sustained signaling model and the limited signaling model (Lever et al., 2014 [\[4\]](#)) (Fig. 1b,c [\[5\]](#)). Both models introduce an additional state to McKeithan's KPR scheme. The sustained signaling model adds an active but ligand-free state P_N^0 , while the limited signaling model introduces an inactivated state I downstream of P_N .

While both models show non-monotonic dependence of signaling activity on ligand affinity (Lever et al., 2014 [\[4\]](#)), only the limited signaling model retains this non-monotonic dependence at saturating ligand concentrations (Lever et al., 2014 [\[4\]](#)) (Fig. 2b [\[6\]](#)), consistent with some paradoxical features observed in RTKs (Freed et al., 2017 [\[7\]](#); Madsen et al., 2025 [\[8\]](#); Myers et al., 2023 [\[9\]](#)). However, the limited signaling model fails to reproduce a second key observation in RTKs: receptor activity in this model increases monotonically with kinase activity, whereas RTK experiments show that partial kinase inhibition can paradoxically increase receptor activity (Kiyatkin et al., 2020 [\[10\]](#); Kleiman et al., 2011 [\[11\]](#)) (Fig. 2c [\[12\]](#)). Thus, these models are insufficient to explain RTK signaling dynamics.

Notably, these models neglect a key feature of many receptor signaling pathways; preferential degradation of activated receptors (Sorkin and Goh, 2009 [\[13\]](#); Koenig and Edwardson, 1997 [\[14\]](#); Charpentier and King, 2021 [\[15\]](#); Cendrowski et al., 2016 [\[16\]](#)). Below, we incorporate preferential degradation in our model to investigate how it governs receptor activity.

A kinetic sorting model integrates active receptor degradation

We build a model to study the effect of two widespread signaling motifs: sequential multi-site phosphorylation and ligand-induced receptor degradation (Fig. 3 [\[17\]](#)) on ligand discrimination. In our model, receptors are delivered to the surface at a constant rate, internalized at a basal rate k_{int} , and degraded more rapidly when highly phosphorylated ($k_{int}^* > k_{int}$). Ligand-bound receptors undergo irreversible phosphorylation and dephosphorylation through distinct irreversible mechanisms. We note that both kinase and phosphatase are irreversible reactions carried out by

Figure 1.

Reaction scheme of kinetic proofreading models.

Chemical species and rate constants are shown in the figure. R denote ligand-free receptors, B denote ligand-bound inactive receptors, and P_n , $n \in [1, N]$ are phosphorylated receptors. The ultimate phosphorylated species P_N (marked red) is assumed to be signaling competent. (a) shows the traditional model first proposed by McKeithan (McKeithan, 1995). (b, c) show the sustained signaling model and the limited signaling model (Lever et al., 2014) which introduce additional receptor states, P_N^0 and I respectively, directly following receptor activation.

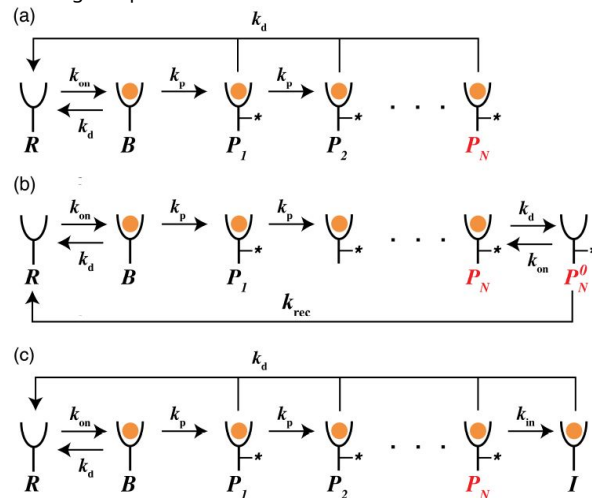
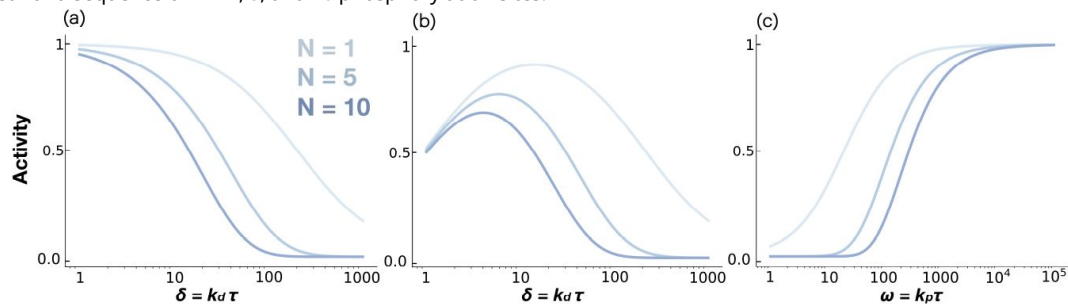


Figure 2.

Ligand discrimination in kinetic proofreading models.

(a) Activity P_N plotted as a function of non-dimensional ligand dissociation rate δ for the traditional KPR scheme (Fig. 1a). (b) Activity P_N plotted as a function of non-dimensional ligand dissociation rate δ for the limited signaling model (Fig. 1b). (c) The dependence of the activity on the dimensionless phosphorylation rate ω for the limited signaling model. All figures plotted for a sequence of $N = 1, 5$, and 10 phosphorylation sites.



separate enzymes. While their effect on the coarse-grained model of the receptor may appear reversible, it is important to note that receptor phosphorylation via ATP hydrolysis and removal of the phosphate group from the receptor corresponds to a futile cycle that does not recharge the ADP molecule to an ATP molecule. In addition to the previously defined dimensionless parameters, we define the dimensionless active receptor degradation rate, $\beta = k_{\text{int}}^a/k_{\text{int}}$, and the relative rate of dephosphorylation, $\rho = k_{\text{dp}}/k_p$. A key feature of our model is that *all* phosphorylated species are signaling competent. Indeed, in many signaling pathways all phosphorylation sites on the receptor (Schulze et al., 2005; Tobin, 2008; Kollwe et al., 2004; Lemmon and Schlessinger, 2010; Latorraca et al., 2020) have downstream effects. Therefore, we define the net activity A_n of phosphorylation site n as all receptor states where the site n is phosphorylated: $A_n = \sum_{m \geq n} P_m$.

Parameter ranges

To ensure that the phenomena captured by our model are relevant to real signaling networks, we selected ranges for the dimensionless parameters based on direct experimental measurements and model fits. Importantly, many of these kinetic processes have comparable rates across diverse receptor systems (Koenig and Edwardson, 1997; Subtil et al., 1994; Liu et al., 2000). Specifically, basal receptor internalization occurs at rates of $k_{\text{int}} \approx 10^{-4}$ – 10^{-3} , s^{-1} (Wiley, 2003), while active receptor internalization is typically faster, at $k_{\text{int}}^a \approx 10^{-3}$ – 10^{-2} , s^{-1} (Wiley, 2003; Lyashenko et al., 2020). Ligand dissociation rates typically fall in the range $k_d \approx 10^{-2}$ – 10^{-1} , s^{-1} (Chen et al., 2009; Lyashenko et al., 2020), and receptor phosphorylation (k_p) and dephosphorylation (k_{dp}) occur at $\sim 10^{-1}$ – 10^0 , s^{-1} (Kleiman et al., 2011; Chen et al., 2009; Lyashenko et al., 2020). For EGFR, equilibrium dissociation constants range from ~ 0.1 , nM for the high-affinity ligand Beta-cellulin to ~ 25 , nM for the low-affinity ligand AREG (Hu et al., 2022; Macdonald-Obermann and Pike, 2014). Based on these values, we set the following ranges for dimensionless parameters: $\beta = k_{\text{int}}^a/k_{\text{int}} \in [1, 100]$, $\rho = k_{\text{dp}}/k_p \in [0.01, 100]$, $\omega = k_{\text{dp}}/k_{\text{int}} \in [1, 1000]$, and $\delta = k_d/k_{\text{int}} \in [1, 1000]$. Finally, the number of phosphorylation sites with known functional roles typically ranges from 5 to 25 (Schulze et al., 2005). These broad ranges comfortably encompass experimentally measured estimates. Unless otherwise specified, our default parameter values are $\delta = 20$, $\omega = 200$, $\rho = 0.01$, $\beta = 50$, and $N = 10$.

Before examining how phosphorylation levels depend on model parameters, we illustrate the mechanism of kinetic sorting of receptor states, which tunes ligand specificity beyond pure thermo-dynamic preference, using a simple example. To that end, we consider a signaling network with $N = 5$ phosphorylation sites interacting with three ligands of distinct affinities—high, medium, and low. We assume the dissociation rates for these ligands are $\delta_H = 20$, $\delta_M = 200$, and $\delta_L = 1000$, respectively. In order to compare our model with aforementioned paradoxical experimental observations which have been performed at saturating ligand concentration, we take the limit $u \rightarrow \infty$.

Fig. 4 shows that low affinity ligands ($\delta_L = 1000$) predominantly sort receptors towards the inactive state B and early phosphorylation states P_n , $n \sim 1$ as frequent ligand unbinding prevents progression to later phosphorylation states. This behavior resembles the traditional KPR mechanism described by McKeithan (McKeithan, 1995). In contrast, receptors bound to high affinity ligands are sorted toward later phosphorylation states, which mark them for enhanced degradation. Here, similar to traditional KPR, the fraction of receptors reaching the final phosphorylation state is highest for high affinity ligands. Yet, the overall receptor pool is reduced due to ligand-induced degradation, lowering net phosphorylation activity. Strikingly, receptors bound to intermediate affinity ligands ($\delta_M = 200$) are sorted towards intermediate phosphorylation states, resulting in maximal phosphorylation output. Below, we show how kinetic parameters govern the ability of the network to overcome thermodynamic preference and acquire ligand specificity.

Figure 3.

Reaction scheme of kinetic sorting model.

Chemical species and rate constants are shown in the figure. R denote ligand-free receptors, B denote ligand-bound inactive receptors, and P_n , $n \in [1, N]$ are phosphorylated receptors. Φ represents an implicit source and sink, corresponding to receptor delivery and internalization, respectively. It does not denote a physical chemical species.

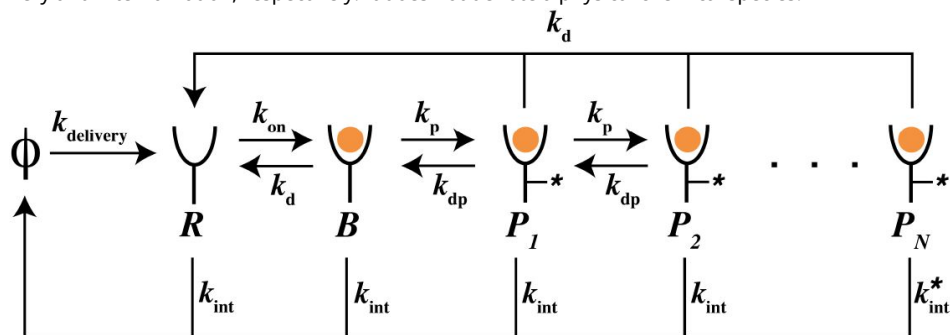
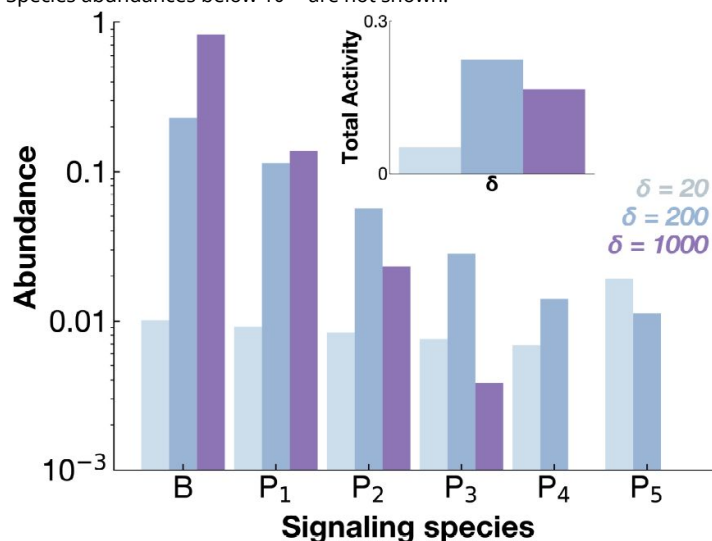


Figure 4.

Kinetic sorting of receptor species.

Abundances of network species B (ligand bound inactive receptor) and P_n , $n \in [1, 5]$ for a signaling receptor with $N = 5$ phosphorylation sites. Abundances are shown for ligands of three different affinities. The inset shows the activity of the first phosphorylation site A_1 . Species abundances below 10^{-3} are not shown.



Early phosphorylation sites show ligand-specificity

Figure 5a illustrates how total phosphorylation activity at each site, A_n , $n \in [1, N]$ varies with ligand dissociation rate δ . We note that the activity of the n^{th} site is given by the total concentration of all species that have the n^{th} site phosphorylated; $A_n = \sum_{i=n}^N p_i$. We find that early phosphorylation sites ($n \sim 1$) exhibit maximal activity at intermediate values of δ while both high- and low-affinity ligands suppress net receptor phosphorylation. Our model predicts that this ligand-specificity diminishes for later sites, where outputs increasingly resemble traditional KPR which favors high-affinity ligands.

To examine how model parameters shape ligand specificity, we focused on the activity at the first phosphorylation site, A_1 , which exhibits the strongest discriminatory behavior (**Fig.5a**). As shown in **Fig.5b**, achieving ligand specificity at high dissociation rates δ requires sufficiently high phosphorylation rates ω . Notably, our model captures a puzzling observation from EGFR signaling: the high-affinity ligand EGF produces lower/comparable steady-state phosphorylation compared to lower-affinity ligands such as Epigen and Epiregulin (Freed et al., 2017; Myers et al., 2023; Madsen et al., 2025). Experimental estimates place the basal EGFR internalization rate at $k_{\text{int}} \approx 1.3 \times 10^{-3} \text{ s}^{-1}$ (Chen et al., 2009), the EGF dissociation rate at $k_d \approx 3 \times 10^{-2} \text{ s}^{-1}$ (Chen et al., 2009), and the phosphorylation rate at $k_p \approx 10^{-1} - 10^0 \text{ s}^{-1}$, yielding $\delta_{\text{EGF}} \approx 10\text{--}20$ and $\omega_{\text{EGFR}} \approx 100\text{--}1000$. Low-affinity ligands such as Epigen (EPGN) and Epiregulin (EREG) have equilibrium dissociation constants about 10-fold higher than EGF (Hu et al., 2022), corresponding to $\delta_{\text{EPGN}} \approx \delta_{\text{EREG}} \approx 100\text{--}200$. The effective degradation rate of fully activated receptors is estimated to be 10–50 times higher than that of inactive receptors (Lyashenko et al., 2020), implying $\beta = 50$. Under these conditions, our model predicts a switch in phosphorylation levels: as δ increases from δ_{EGF} to δ_{EPGN} , receptor phosphorylation increases—reversing the expectation based purely on thermodynamic affinity. This effect arises because EGF-bound receptors are efficiently sorted towards degradation-prone states compared to those bound to lower-affinity ligands.

Our model also explains another paradox in EGFR signaling. Experimental studies have shown that EGF-stimulated receptors exhibit higher steady-state phosphorylation when kinase activity is partially inhibited (Kiyatkin et al., 2020; Kleiman et al., 2011). As shown in **Fig.5c**, at low δ values (e.g., $\delta = 16$), decreasing the phosphorylation rate ω from levels typical of EGFR ($\omega_{\text{EGFR}} \approx 100\text{--}1000$) paradoxically increases overall receptor phosphorylation. A similar effect is observed when receptor dephosphorylation is enhanced (**Fig.5d**). Importantly, our model makes a testable prediction: the reversal of thermodynamic preference observed between EGF and EPGN/EREG will disappear when kinase activity is mildly suppressed (see, e.g., the curves for $\omega = 256$ and $\omega = 16$ over $\delta \in [10, 100]$), such as by treatment with low doses of the kinase inhibitor gefitinib (Herbst et al., 2004). This non-monotonic trend may help prevent cells with abnormally high kinase activity from becoming constitutively active, thereby preserving their sensitivity to extracellular cues.

Multi-site phosphorylation and ligand-induced degradation are both essential for ligand-specificity

To assess the importance of sequential multi-site phosphorylation on ligand specificity, we analyzed A_1^N , the phosphorylation of the first site for signaling networks with N phosphorylation sites. **Fig. 6a** shows that multi-site phosphorylation is essential to endow signaling networks with ligand specificity and ligand-induced receptor degradation alone is not sufficient. This is because the non-monotonic preference for intermediate affinity ligands arises only when the receptors can be sorted among multiple phosphorylation sites: earlier ones for low affinity ligands and later ones for high affinity ligands.

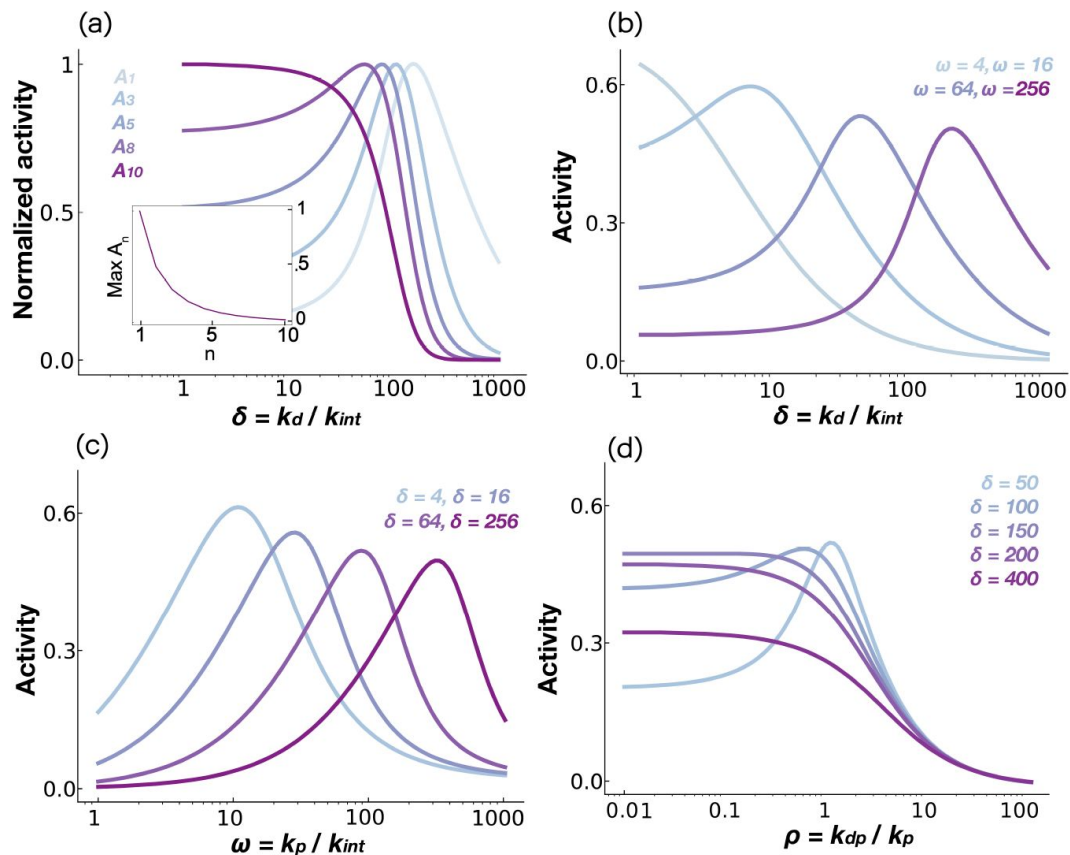


Figure 5.

Kinetic sorting model predicts ligand specificity.

(a) The activity A_n of the n^{th} phosphorylation site as a function of dimensionless dissociation rate δ . The activity is normalized to the maximum activity. The maximum A_n as a function of n is shown in the inset. (b) Activity of the first phosphorylation site A_1 plotted as a function of the dissociation rate δ for different values of the phosphorylation rate ω . (c, d) Activity of the first phosphorylation site A_1 plotted as a function of phosphorylation rate ω (dephosphorylation rate ρ in panel d) for different values of the dissociation rate δ .

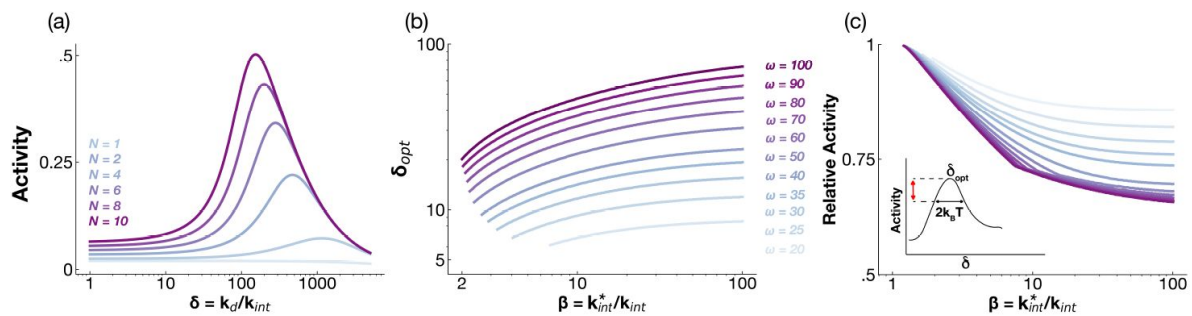


Figure 6.

Multiple phosphorylation sites and receptor degradation dictate ligand specificity.

(a) Activity of the first phosphorylation site, A_1 , as a function of the dissociation rate δ for signaling networks with different number of phosphorylation sites. (b) The optimal dissociation rate δ_{opt} that leads to maximum phosphorylation activity as a function of dimensionless degradation rate β for different values of ω . δ_{opt} is shown only if $\delta_{opt} \in [1, 1000]$. (c) The relative activity of a ligand with dissociation rate that differs by $k_B T$ compared to δ_{opt} plotted as a function of β for different values of ω (see inset). Of the two ligands that differ in stability by $k_B T$, the ligand exhibiting maximum activity is considered.

To assess how receptor degradation shapes ligand specificity for a multi-site phosphorylation network, we examined how altering receptor turnover influences model behavior. As shown in **Fig. 6b**, the optimal dissociation rate δ_{opt} , which maximizes receptor phosphorylation levels, increases with ligand-induced degradation rate β . Crucially, this optimal δ_{opt} emerges only when receptor degradation is strong ($\beta \gg 1$). These predictions can be tested by blocking receptor degradation, e.g., via mutation of ubiquitination sites (Gerritsen et al., 2023).

To quantify ligand specificity, we computed receptor phosphorylation in response to ligands differing by at least one $k_B T$ in binding free energy from the optimal ligand. **Figure 6c** shows that as β increases, phosphorylation downstream of suboptimal ligands (red line in inset) declines relative to the optimal ligand. This enhanced specificity is further amplified by increasing kinase activity ω .

These results show that both multi-site phosphorylation and ligand induced degradation are key features controlling ligand specificity in our kinetic sorting mechanism.

Discussion

Cells face the formidable task of decoding multiple chemically distinct extracellular signals to generate appropriate, context-specific responses. This challenge is especially acute for cell surface receptors like RTKs, GPCRs, and interleukin receptors, which bind multiple cognate ligands and yet elicit distinct downstream outcomes. While equilibrium affinity provides a baseline expectation for ligand specificity, it cannot fully explain the rich and often counterintuitive behaviors observed in many signaling systems.

Here, we show that a non-equilibrium mechanism of *kinetic sorting* which operates through multi-site phosphorylation and active receptor degradation can explain how signaling networks achieve ligand specificity beyond equilibrium limits. In kinetic sorting, high-affinity ligand-receptor complexes are sorted toward degradation-prone states, low-affinity complexes are sorted towards inactivated states, and intermediate-affinity ligands strike the optimal balance between progression and degradation to maximize signaling. This framework explains paradoxical features observed in RTK systems, including the non-monotonic dependence of phosphorylation on ligand affinity and kinase activity. Importantly, our model predicts that early phosphorylation sites show the strongest ligand discrimination, consistent with recent experimental observations. It also makes the testable prediction that impairing receptor degradation should reduce specificity by eliminating the kinetic sorting effect. Given the ubiquity of the essential motifs of our mechanism, i.e., multi-site phosphorylation and receptor degradation, we believe that kinetic sorting may be a common mechanism to modulate ligand specificity at the receptor level, potentially in addition to other mechanisms that endow signaling networks with ligand-specificity, both at the receptor level (Lever et al., 2014) as well as in downstream signaling pathways (Singh and Nemenman, 2017).

In contrast to what has been shown previously for KPR models (Coombs et al., 2002; Lever et al., 2014), the kinetic sorting model also captures the non-monotonic relationship between signaling output and kinase/phosphatase activity observed in RTK systems such as EGFR (Kleiman et al., 2011; Kiyatkin et al., 2020). In these systems, partial inhibition of kinase activity paradoxically increases steady-state receptor phosphorylation, a behavior not accounted for by equilibrium models (see Supplementary Materials) or by prior non-equilibrium schemes such as the limited signaling model (Lever et al., 2014). This type of protective filtering can ensure that downstream signaling remains contingent on extracellular cues and is not constitutively active, thereby preventing persistent, cue-independent activation. Such regulation could help maintain control in pathways such as those governing growth, where deregulated activity can have severe consequences. The potential benefit of this regulatory pattern suggests it could be advantageous in

other signaling contexts. Consistent with this idea, non-monotonic regulation by kinase or phosphatase activity is found in other systems through distinct mechanisms (e.g., the non-monotonic effects of the phosphatase CD45 in T-cell receptor signaling (Courtney et al., 2019)). This indicates that selective filtering based on enzymatic activity is a strategy employed in diverse biological settings. While direct evidence for the kinetic sorting mechanism remains limited to RTKs, similar filtering behavior emerges in theoretical analyses of phosphorylation–dephosphorylation cycles in more general settings (Martins and Swain, 2013), suggesting it may represent a broader principle of enzymatic signaling networks.

Our findings complement prior studies on mechanisms of ligand specificity that operate at thermal equilibrium, such as those described in the Bone Morphogenetic Protein (BMP) pathway (Antebi et al., 2017; Su et al., 2022; Parres-Gold et al., 2025). BMP signaling relies on promiscuous ligand–receptor interactions, with specificity emerging from differences in receptor abundance, binding affinity, and complex activity. In contrast, our work shows that non-equilibrium mechanisms—such as phosphorylation cycles and ligand-induced receptor degradation—can achieve ligand discrimination even for a single receptor type. Given that ligand–receptor promiscuity, multisite phosphorylation, and receptor turnover are common features across signaling systems (e.g., in the EGFR/ErbB family (Linggi and Carpenter, 2006)), it is likely that biological networks integrate both equilibrium and non-equilibrium strategies to achieve robust and tunable ligand specificity.

In recent years, there has been growing interest in engineering synthetic physical and chemical circuits capable of carrying out complex computational tasks, including input discrimination, classification, prediction, and the generation of multiple stable cell states (Shakiba et al., 2021; Ma et al., 2022; Benzinger et al., 2022; Zhu et al., 2022; Floyd et al., 2024; Parres-Gold et al., 2025; Aoki et al., 2019). Some of these synthetic strategies rely on equilibrium thermodynamics (Parres-Gold et al., 2025), while others exploit non-equilibrium steady states (Floyd et al., 2024). We propose that non-equilibrium kinetic sorting, which harnesses receptor synthesis and degradation, could provide synthetic biologists with a powerful framework for achieving precise control over molecular abundances and dynamic system behavior.

Finally, we address a major concern in non-equilibrium signaling circuits: the energetic cost of operation. Previous theoretical work has shown that free energy dissipation places fundamental constraints on the performance of signaling networks (Bryant and Machta, 2023; Govern and ten Wolde, 2014; Lan et al., 2012; Mehta and Schwab, 2012; Qian and Reluga, 2005; Cao et al., 2015; Azeloglu and Iyengar, 2015; Floyd et al., 2024; Mahdavi et al., 2024). These studies typically focus on futile cycles of reversible modifications such as phosphorylation or methylation. In contrast, ligand-induced receptor degradation—a central feature of many signaling networks—is a far more energy-intensive process. For example, MCF10A cells maintain approximately 10^5 EGFR molecules on the surface (each 1,210 amino acids in length) (Shi et al., 2016), with a synthesis rate of about 15 receptors per second (Lyashenko et al., 2020), corresponding to an energetic cost of roughly $\sim 8 \times 10^4$ ATP/sec (assuming 4.5 ATP per peptide bond (Milo et al., 2010)). By comparison, EGFR dephosphorylation occurs over ~ 15 seconds (Kleiman et al., 2011), and only 5 – 10% of receptors are phosphorylated at steady state (Shi et al., 2016; Feng et al., 2023), resulting in a much lower energetic cost of $\sim 6 \times 10^2$ ATP/sec for dephosphorylation. Thus, the energetic burden of receptor turnover can exceed that of reversible modification cycles by up to two orders of magnitude. These estimates suggest that, at least in eukaryotic cells where signaling proteins may turnover multiple times within cellular lifetime (Milo et al., 2010), non-equilibrium modification cycles are unlikely to pose a fundamental energetic limitation on the functionality of signaling networks. Here, the energetic demands of signaling networks must account for protein turnover in addition to non-equilibrium modification cycles.

Supplementary materials

Equations for proofreading models

The equations describing species abundances in the traditional KPR model similar to that of McKeithan (McKeithan, 1995 [\[1\]](#)) are as follows:

$$\frac{dR}{dt} = -k_{on}LR + k_dB + k_d \sum_{i=1}^N P_i \quad (2)$$

$$\frac{dB}{dt} = +k_{on}LR - k_dB - k_pB \quad (3)$$

$$\frac{dP_1}{dt} = k_pB - k_pP_1 - k_dP_1 \quad (4)$$

$$\frac{dP_i}{dt} = k_pP_{i-1} - k_pP_i - k_dP_i \quad \forall i \in [2, N-1] \quad (5)$$

$$\frac{dP_N}{dt} = k_pP_{N-1} - k_dP_N \quad (6)$$

For the limited signaling model, the dynamics of B , and P_i , $i \in [1, N-1]$ are identical to the traditional KPR model. The dynamics of R and P_N are modified as follows:

$$\frac{dR}{dt} = -k_{on}LR + k_dB + k_d \sum_{i=1}^N P_i + k_dI \quad (7)$$

$$\frac{dP_N}{dt} = k_pP_{N-1} - k_dP_N - k_{in}P_N \quad (8)$$

$$\frac{dI}{dt} = k_{in}P_N - k_dP_I \quad (9)$$

Equations for the model with receptor degradation

Signaling receptors participate in a variety of complex regulatory processes, including non-linear ligand binding dynamics (Limbird et al., 1975 [\[2\]](#); Macdonald and Pike, 2008 [\[3\]](#)), receptor oligomerization (Mudumbi et al., 2024 [\[4\]](#); Huang et al., 2016 [\[5\]](#)), context-specific interactions with adapter proteins (Madsen and Vanhaesebroeck, 2020 [\[6\]](#); Feng et al., 2023 [\[7\]](#)), and trafficking between cellular compartments leading to degradation (Sorkin and Goh, 2009 [\[8\]](#); Wiley, 2003 [\[9\]](#); Irannejad and Von Zastrow, 2014 [\[10\]](#)).

While computational models that incorporate these mechanistic details are powerful tools for hypothesis generation (Chen et al., 2010 [\[11\]](#); Qiao et al., 2025 [\[12\]](#)), they often require large-scale datasets for accurate parameterization (Feng et al., 2023 [\[7\]](#)). As an alternative, simplified models that intentionally omit certain mechanistic details can still yield deep qualitative insights, even if they cannot quantitatively reproduce experimental data.

In this study, we present such a simplified model aimed at explaining two paradoxical features of receptor tyrosine kinase (RTK) signaling: (1) the non-monotonic relationship between ligandreceptor affinity and steady-state receptor phosphorylation (Freed et al., 2017 [\[13\]](#); Madsen et al., 2025 [\[14\]](#); Myers et al., 2023 [\[15\]](#)), and (2) the counterintuitive increase in receptor phosphorylation following mild kinase inhibition (Kleiman et al., 2011 [\[16\]](#); Kiyatkin et al., 2020 [\[17\]](#)).

To keep the model simple and tractable, we neglect receptor recycling and oligomerization. Previously, we showed that the combined effects of endocytosis, recycling, and degradation can be captured by a single effective dimensionless parameter, β in this study, which reflects the degradation bias of fully phosphorylated receptors compared to partially phosphorylated receptors (Lyashenko et al., 2020). Similarly, receptor dimerization and negative cooperativity can be abstracted into a Hill coefficient $\eta < 1$ (Lyashenko et al., 2020). For the phenomena explored here, including oligomerization would modify the shape of the response curves but not their qualitative behavior.

Under these assumptions, the governing equations for the model are given by:

$$\frac{dR}{dt} = k_{\text{delivery}} - k_{\text{on}}LR + k_dB + k_d \sum_{i=1}^N P_i - k_{\text{int}}R \quad (10)$$

$$\frac{dB}{dt} = k_{\text{on}}LR - k_dB - k_pB - k_{\text{int}}B \quad (11)$$

$$\frac{dP_1}{dt} = k_pB - k_pP_1 - k_dP_1 - k_{\text{int}}P_1 \quad (12)$$

$$\begin{aligned} \frac{dP_i}{dt} &= k_pP_{i-1} - k_pP_i - k_dP_i \\ &- k_{\text{int}}P_i \quad \forall i \in [2, N-1] \end{aligned} \quad (13)$$

$$\frac{dP_N}{dt} = k_pP_{N-1} - k_dP_N - k_{\text{int}}^*P_N \quad (14)$$

All equations are solved at steady state and in the limit $u \rightarrow \infty$. All codes required to generate the figures in the manuscript can be found at <https://github.com/BarriosJer0/KineticSorting>.

Equations for a model at thermal equilibrium

To confirm the role of non-equilibrium thermodynamics on ligand specificity, we consider the closest equivalent equilibrium model. The strongest requirement of an equilibrium model is that all reactions must be bidirectional. Another requirement is that microscopic reversibility or *detailed balance*. Specifically, ratios of rate constants around loops must equal to unity for all loops. The first requirement implies that unidirectional reactions: synthesis and degradation of receptors and the irreversible loss of activity due to ligand dissociation cannot exist in a reaction network that operates at equilibrium. The simplest equilibrium model closest to the kinetic sorting scheme is governed by the following equations:

$$\frac{dR}{dt} = -k_{\text{on}}LR + k_dB \quad (15)$$

$$\frac{dB}{dt} = k_{\text{on}}LR - k_dB - k_pB + k_{\text{dp}}P_1 \quad (16)$$

$$\frac{dP_i}{dt} = k_pP_{i-1} - k_{\text{dp}}P_i - k_pP_i + k_{\text{dp}}P_{i+1} \quad \forall i \in [1, N-1] \quad (17)$$

$$\frac{dP_N}{dt} = k_pP_{N-1} - k_{\text{dp}}P_N \quad (18)$$

In the above equations, we use the notation $P_0 \equiv B$.

We note that phosphorylation/dephosphorylation reactions are unidirectional non-equilibrium reactions carried out by different enzymes: Phosphorylation hydrolyses ATP to ADP and attaches a phosphate group to the receptor. In contrast, while dephosphorylation removes phosphate group from the receptor, it does not recharge an ADP molecule back to ATP. Notably, however, this non-equilibrium nature of the phosphorylation/dephosphorylation cycle is not apparent in our coarse-grained kinetic scheme where ATP and ADP are not explicitly considered. We retain this part of the non-equilibrium model since a corresponding equilibrium model can be imagined where different sites on the receptor change conformation between an inactive and an active state and that these changes occur in a sequential manner.

Solving these equations at steady state and taking the limit $u = Lk_{\text{on}}/k_{\text{d}} \rightarrow \infty$, we have

$$p_i = \frac{P_i}{R_T} = \frac{\rho^{N-i}}{\sum_{i=0}^N \rho^i} \quad (19)$$

where R_T is the total number of receptors and $\rho = k_{\text{dp}}/k_{\text{p}}$. Note that as expected, this equilibrium model has no dependence on ligand dissociation rate k_{d} at saturation, further confirming that non-equilibrium reactions are needed to endow cells with ligand specificity.

Additional information

Funding

National Institutes of Health (R35GM142547)

Sir Henry Wellcome Fellowship (220464/Z/20/Z)

References

1. Antebi YE, Linton JM, Klumpe H, Bintu B, Gong M, Su C, McCardell R, Elowitz MB (2017) **Combinatorial signal perception in the BMP pathway** *Cell* **170**:1184–1196 [Google Scholar](#)
2. Aoki SK, Lillacci G, Gupta A, Baumschlager A, Schweingruber D, Khammash M. (2019) **A universal biomolecular integral feedback controller for robust perfect adaptation** *Nature* **570**:533–537 [Google Scholar](#)
3. Azeloglu EU, Iyengar R. (2015) **Signaling networks: information flow, computation, and decision making** *Cold Spring Harbor perspectives in biology* **7**:a005934 [Google Scholar](#)
4. Benzinger D, Ovinnikov S, Khammash M. (2022) **Synthetic gene networks recapitulate dynamic signal decoding and differential gene expression** *Cell Systems* **13**:353–364 [Google Scholar](#)
5. Bryant SJ, Machta BB (2023) **Physical constraints in intracellular signaling: The cost of sending a bit** *Physical review letters* **131**:068401 [Google Scholar](#)
6. Cantley LC, Hunter T, Sever R, Thorner JW (2014) **Signal transduction: principles, pathways, and processes** Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press [Google Scholar](#)
7. Cao Y, Wang H, Ouyang Q, Tu Y. (2015) **The free-energy cost of accurate biochemical oscillations** *Nature physics* **11**:772–778 [Google Scholar](#)
8. Cendrowski J, Mamińska A, Miaczynska M. (2016) **Endocytic regulation of cytokine receptor signaling** *Cytokine & growth factor reviews* **32**:63–73 [Google Scholar](#)
9. Charpentier JC, King PD (2021) **Mechanisms and functions of endocytosis in T cells** *Cell Communication and Signaling* **19**:92 [Google Scholar](#)
10. Chen WW, Niepel M, Sorger PK (2010) **Classic and contemporary approaches to modeling biochemical reactions** *Genes & development* **24**:1861–1875 [Google Scholar](#)
11. Chen WW, Schoeberl B, Jasper PJ, Niepel M, Nielsen UB, Lauffenburger DA, Sorger PK (2009) **Input-output behavior of ErbB signaling pathways as revealed by a mass action model trained against dynamic data** *Molecular systems biology* **5**:239 [Google Scholar](#)
12. Clark RB, Knoll BJ, Barber R. (1999) **Partial agonists and G protein-coupled receptor desensitization** *Trends in pharmacological sciences* **20**:279–286 [Google Scholar](#)
13. Coombs D, Kalergis AM, Nathenson SG, Wofsy C, Goldstein B. (2002) **Activated TCRs remain marked for internalization after dissociation from pMHC** *Nature immunology* **3**:926–931 [Google Scholar](#)
14. Courtney AH, Shvets AA, Lu W, Griffante G, Mollenauer M, Horkova V, Lo WL, Yu S, Stepanek O, Chakraborty AK, Weiss A. (2019) **CD45 functions as a signaling gatekeeper in T cells** *Science Signaling* **12**:eaaw8151 [Google Scholar](#)

15. Feng S, Sanford JA, Weber T, Hutchinson-Bunch CM, Dakup PP, Paurus VL, Attah K, Sauro HM, Qian WJ, Wiley HS (2023) **A Phosphoproteomics Data Resource for Systems-level Modeling of Kinase Signaling Networks** *bioRxiv* [Google Scholar](#)
16. Floyd C, Dinner AR, Murugan A, Vaikuntanathan S. (2024) **Limits on the computational expressivity of non-equilibrium biophysical processes** *arXiv* [Google Scholar](#)
17. Freed DM, Bessman NJ, Kiyatkin A, Salazar-Cavazos E, Byrne PO, Moore JO, Valley CC, Ferguson KM, Leahy DJ, Lidke DS, et al. (2017) **EGFR ligands differentially stabilize receptor dimers to specify signaling kinetics** *Cell* **171**:683–695 [Google Scholar](#)
18. Gerritsen JS, Faraguna JS, Bonavia R, Furnari FB, White FM (2023) **Predictive data-driven modeling of C-terminal tyrosine function in the EGFR signaling network** *Life Science Alliance* **6** [Google Scholar](#)
19. Govern CC, ten Wolde PR (2014) **Energy dissipation and noise correlations in biochemical sensing** *Physical review letters* **113**:258102 [Google Scholar](#)
20. Gulukota K, Wolynes PG (1994) **Statistical mechanics of kinetic proofreading in protein folding in vivo** *Proceedings of the National Academy of Sciences* **91**:9292–9296 [Google Scholar](#)
21. Herbst RS, Fukuoka M, Baselga J. (2004) **Gefitinib—a novel targeted approach to treating cancer** *Nature Reviews Cancer* **4**:956–965 [Google Scholar](#)
22. Hilleren P, Parker R. (1999) **mRNA surveillance in eukaryotes: kinetic proofreading of proper translation termination as assessed by mRNP domain organization?** *Rna* **5**:711–719 [Google Scholar](#)
23. Hopfield JJ (1974) **Kinetic proofreading: a new mechanism for reducing errors in biosynthetic processes requiring high specificity** *Proceedings of the National Academy of Sciences* **71**:4135–4139 [Google Scholar](#)
24. Hopfield JJ (1980) **The energy relay: a proofreading scheme based on dynamic cooperativity and lacking all characteristic symptoms of kinetic proofreading in DNA replication and protein synthesis** *Proceedings of the National Academy of Sciences* **77**:5248–5252 [Google Scholar](#)
25. Hu C, Leche CA, Kiyatkin A, Yu Z, Stayrook SE, Ferguson KM, Lemmon MA (2022) **Glioblastoma mutations alter EGFR dimer structure to prevent ligand bias** *Nature* **602**:518–522 [Google Scholar](#)
26. Huang WY, Alvarez S, Kondo Y, Lee YK, Chung JK, Lam HYM, Biswas KH, Kuriyan J, Groves JT (2019) **A molecular assembly phase transition and kinetic proofreading modulate Ras activation by SOS** *Science* **363**:1098–1103 [Google Scholar](#)
27. Huang Y, Bharill S, Karandur D, Peterson SM, Marita M, Shi X, Kaliszewski MJ, Smith AW, Isacoff EY, Kuriyan J. (2016) **Molecular basis for multimerization in the activation of the epidermal growth factor receptor** *eLife* **5**:e14107 <https://doi.org/10.7554/eLife.14107> | [Google Scholar](#)
28. Irannejad R, Von Zastrow M. (2014) **GPCR signaling along the endocytic pathway** *Current opinion in cell biology* **27**:109–116 [Google Scholar](#)
29. Kiyatkin A, van Alderwerelt, van Rosenburgh IK, Klein DE, Lemmon MA (2020) **Kinetics of receptor tyrosine kinase activation define ERK signaling dynamics** *Science Signaling*

13:eaaz5267 [Google Scholar](#)

30. Kleiman LB, Maiwald T, Conzelmann H, Lauffenburger DA, Sorger PK (2011) **Rapid phospho-turnover by receptor tyrosine kinases impacts downstream signaling and drug binding** *Molecular cell* **43**:723–737 [Google Scholar](#)
31. Koenig JA, Edwardson JM (1997) **Endocytosis and recycling of G protein-coupled receptors** *Trends in pharmacological sciences* **18**:276–287 [Google Scholar](#)
32. Kollwe C, Mackensen AC, Neumann D, Knop J, Cao P, Li S, Wesche H, Martin MU (2004) **Sequential autophosphorylation steps in the interleukin-1 receptor-associated kinase-1 regulate its availability as an adapter in interleukin-1 signaling** *Journal of Biological Chemistry* **279**:5227–5236 [Google Scholar](#)
33. Lan G, Sartori P, Neumann S, Sourjik V, Tu Y. (2012) **The energy-speed-accuracy trade-off in sensory adaptation** *Nature physics* **8**:422–428 [Google Scholar](#)
34. Latorraca NR, Masureel M, Hollingsworth SA, Heydenreich FM, Suomivuori CM, Brinton C, Townshend RJ, Bouvier M, Kobilka BK, Dror RO (2020) **How GPCR phosphorylation patterns orchestrate arrestin-mediated signaling** *Cell* **183**:1813–1825 [Google Scholar](#)
35. Lemmon MA, Schlessinger J. (2010) **Cell signaling by receptor tyrosine kinases** *Cell* **141**:1117–1134 [Google Scholar](#)
36. Lever M, Maini PK, Van Der Merwe PA, Dushek O. (2014) **Phenotypic models of T cell activation** *Nature Reviews Immunology* **14**:619–629 [Google Scholar](#)
37. Limbird LE, De Meyts P, Lefkowitz RJ (1975) **β -Adrenergic receptors: evidence for negative cooperativity** *Biochemical and biophysical research communications* **64**:1160–1168 [Google Scholar](#)
38. Linggi B, Carpenter G. (2006) **ErbB receptors: new insights on mechanisms and biology** *Trends in cell biology* **16**:649–656 [Google Scholar](#)
39. Liu H, Rhodes M, Wiest DL, Vignali DA (2000) **On the dynamics of TCR: CD3 complex cell surface expression and downmodulation** *Immunity* **13**:665–675 [Google Scholar](#)
40. Lyashenko E, Niepel M, Dixit PD, Lim SK, Sorger PK, Vitkup D. (2020) **Receptor-based mechanism of relative sensing and cell memory in mammalian signaling networks** *eLife* **9**:e50342 <https://doi.org/10.7554/eLife.50342> | [Google Scholar](#)
41. Ma Y, Budde MW, Mayalu MN, Zhu J, Lu AC, Murray RM, Elowitz MB (2022) **Synthetic mammalian signaling circuits for robust cell population control** *Cell* **185**:967–979 [Google Scholar](#)
42. Macdonald JL, Pike LJ (2008) **Heterogeneity in EGF-binding affinities arises from negative cooperativity in an aggregating system** *Proceedings of the National Academy of Sciences* **105**:112–117 [Google Scholar](#)
43. Macdonald-Obermann JL, Pike LJ (2014) **Different epidermal growth factor (EGF) receptor ligands show distinct kinetics and biased or partial agonism for homodimer and heterodimer formation** *Journal of Biological Chemistry* **289**:26178–26188 [Google Scholar](#)

44. Madsen RR, Le Marois A, Mruk ON, Voliotis M, Yin S, Sufi J, Qin X, Zhao SJ, Gorczynska J, Morelli D, et al. (2025) **Oncogenic PIK3CA corrupts growth factor signaling specificity** *Molecular Systems Biology* **21**:126–157 [Google Scholar](#)
45. Madsen RR, Vanhaesebroeck B. (2020) **Cracking the context-specific PI3K signaling code** *Science Signaling* **13**:eaay2940 [Google Scholar](#)
46. Mahdavi SD, Salmon GL, Daghlani P, Garcia HG, Phillips R. (2024) **Flexibility and sensitivity in gene regulation out of equilibrium** *Proceedings of the National Academy of Sciences* **121**:e2411395121 [Google Scholar](#)
47. Martins BMC, Swain PS (2013) **Ultrasensitivity in Phosphorylation-Dephosphorylation Cycles with Little Substrate** *PLoS Computational Biology* **9**:e1003175 [Google Scholar](#)
48. McKeithan TW (1995) **Kinetic proofreading in T-cell receptor signal transduction** *Proceedings of the national academy of sciences* **92**:5042–5046 [Google Scholar](#)
49. Mehta P, Schwab DJ (2012) **Energetic costs of cellular computation** *Proceedings of the National Academy of Sciences* **109**:17978–17982 [Google Scholar](#)
50. Milo R, Jorgensen P, Moran U, Weber G, Springer M. (2010) **BioNumbers—the database of key numbers in molecular and cell biology** *Nucleic acids research* **38**:D750–D753 [Google Scholar](#)
51. Mudumbi KC, Burns EA, Schodt DJ, Petrova ZO, Kiyatkin A, Kim LW, Mangiacapre EM, Ortiz-Caraveo I, Ortiz HR, Hu C, et al. (2024) **Distinct interactions stabilize EGFR dimers and higher-order oligomers in cell membranes** *Cell reports* **43** [Google Scholar](#)
52. Murugan A, Huse DA, Leibler S. (2014) **Discriminatory proofreading regimes in nonequilibrium systems** *Physical Review X* **4**:021016 [Google Scholar](#)
53. Myers PJ, Lee SH, Lazzara MJ (2023) **An integrated mechanistic and data-driven computational model predicts cell responses to high-and low-affinity EGFR ligands** *bioRxiv* [Google Scholar](#)
54. Ninio J. (1975) **Kinetic amplification of enzyme discrimination** *Biochimie* **57**:587–595 [Google Scholar](#)
55. Parres-Gold J, Levine M, Emert B, Stuart A, Elowitz MB (2025) **Contextual computation by competitive protein dimerization networks** *Cell* [Google Scholar](#)
56. Qian H, Reluga TC (2005) **Nonequilibrium thermodynamics and nonlinear kinetics in a cellular signaling switch** *Physical review letters* **94**:028101 [Google Scholar](#)
57. Qiao L, Khalilimeybodi A, Linden-Santangeli NJ, Rangamani P. (2025) **The evolution of systems biology and systems medicine: From mechanistic models to uncertainty quantification** *Annual Review of Biomedical Engineering* **27** [Google Scholar](#)
58. Schulze WX, Deng L, Mann M. (2005) **Phosphotyrosine interactome of the ErbB-receptor kinase family** *Molecular systems biology* **1**:2005–8 [Google Scholar](#)

59. Shakiba N, Jones RD, Weiss R, Del Vecchio D. (2021) **Context-aware synthetic biology by controller design: Engineering the mammalian cell** *Cell systems* **12**:561–592 [Google Scholar](#)
60. Shi T, Niepel M, McDermott JE, Gao Y, Nicora CD, Chrisler WB, Markillie LM, Petyuk VA, Smith RD, Rodland KD, et al. (2016) **Conservation of protein abundance patterns reveals the regulatory architecture of the EGFR-MAPK pathway** *Science Signaling* **9**:rs6–rs6 [Google Scholar](#)
61. Singh V, Nemenman I. (2017) **Simple biochemical networks allow accurate sensing of multiple ligands with a single receptor** *PLoS Computational Biology* **13**:e1005490 [Google Scholar](#)
62. Sorkin A, Goh LK (2009) **Endocytosis and intracellular trafficking of ErbBs** *Experimental cell research* **315**:683–696 [Google Scholar](#)
63. Su CJ, Murugan A, Linton JM, Yeluri A, Bois J, Klumpe H, Langley MA, Antebi YE, Elowitz MB (2022) **Ligand-receptor promiscuity enables cellular addressing** *Cell systems* **13**:408–425 [Google Scholar](#)
64. Subtil A, Hémar A, Dautry-Varsat A. (1994) **Rapid endocytosis of interleukin 2 receptors when clathrin-coated pit endocytosis is inhibited** *Journal of cell science* **107**:3461–3468 [Google Scholar](#)
65. Tobin A. (2008) **G-protein-coupled receptor phosphorylation: Where, when and by whom** *British journal of pharmacology* **153**:S167–S176 [Google Scholar](#)
66. Wiley HS (2003) **Trafficking of the ErbB receptors and its influence on signaling** *Experimental Cell Research* :81–91 [Google Scholar](#)
67. Zhu R, del Rio-Salgado JM, Garcia-Ojalvo J, Elowitz MB (2022) **Synthetic multistability in mammalian cells** *Science* **375**:eabg9765 [Google Scholar](#)

Author information

Andrew Goetz

Department of Biomedical Engineering, Yale University, New Haven, United States

For correspondence: andrew.goetz@yale.edu

Jeremy Barrios

Department of Physics, Yale University, New Haven, United States

Ralitsa Madsen

MRC Protein Phosphorylation and Ubiquitylation Unit, University of Dundee, Dundee, United Kingdom

ORCID iD: [0000-0001-8844-5167](https://orcid.org/0000-0001-8844-5167)

Purushottam Dixit

Department of Biomedical Engineering, Yale University, New Haven, United States, Systems Biology Institute, Yale University, West Haven, United States

For correspondence: purushottam.dixit@yale.edu

Editors

Reviewing Editor

Arvind Murugan

University of Chicago, Chicago, United States of America

Senior Editor

Aleksandra Walczak

CNRS, Paris, France

Reviewer #1 (Public review):

Summary:

The authors study the steady-state solutions of ODE models for molecular signaling involving ligand binding coupled to multi-site phosphorylation at saturating ligand concentrations. Although the results are in principle general, the work highlights the receptor tyrosine kinases (RTK) as model systems. After presenting previous ODE model solutions, the authors present their own "kinetic sorting" model, which is distinguished by ligand-induced phosphorylation-dependent receptor degradation and the property that every phosphorylation state is signaling competent. The authors show that this model recovers the two types of non-monotonicity experimentally reported for RTKs: maximum activity for intermediate ligand affinity and maximum activity for intermediate kinase activity.

The main contribution of the work is in demonstrating that their model can capture both types of non-monotonicity, whereas previous models could at most capture non-monotonicity of ligand binding.

Strengths:

The question of how energy dissipating, and thus non-equilibrium, molecular systems can achieve steady-state solutions not accessible to equilibrium systems is of fundamental importance in biomolecular information processing and self-organization. Although the authors do not address the energy requirements of their non-equilibrium model, their comparative analysis of different alternative non-equilibrium models provides insight into the design choices necessary to achieve non-monotonic control, a property that is inaccessible at equilibrium.

The paper is succinctly written and easy to follow, and the authors achieve their aims by providing convincing numerical solutions demonstrating non-monotonicity over the range of parameter values encompassing the biologically relevant regime.

Weaknesses:

(1) A key motivating framework for this work is the argument that the ability to tune to recognize intermediate ligand affinities provides a control knob for signal selection that is available to non-equilibrium systems. As such, this seems like a compelling type of ligand selectivity, which is a question of broad interest. However, as the authors note in the results, the previously published "limited signaling model" already achieves such non-monotonicity to ligand binding affinity. The introduction and abstract do not clearly delineate the new contributions of the model.

The novel benefit of the model introduced by the authors is that it also achieves non-monotonic response to kinase activity. Because such non-monotonicity is observed for RTK, this would make the authors' model a better fit for capturing RTK behavior. However, the broad significance of achieving non-monotonicity to kinase activity is not motivated or supported by empirical evidence in the paper. As such, the conceptual significance of the modified model presented by the authors is not clear.

UPDATE: The authors have now clarified the significance of the model in elucidating how known motifs (multisite phosphorylation and active receptor degradation) could explain the behavior, including non-monotonicity. The authors have also provided compelling arguments for the biological significance of achieving non-monotonic kinase activity response.

(2) Whereas previous models used in the literature are schematized in Figure 1, the model proposed by the author is missing (See line 97 of page 3). Without the schematic, the text description of the model is incomplete.

UPDATE: this issue has been resolved.

(3) The authors use the activity of the first phosphorylation site as the default measure of activity. This choice needs to be justified. Why not use the sum of the activities at all sites?

UPDATE: This was a non-issue. The potential misunderstanding has been mitigated by clarifications in the text.

Comments on revisions:

All issues previously identified were convincingly addressed. I have no additional suggestions.

<https://doi.org/10.7554/eLife.107524.2.sa2>

Reviewer #2 (Public review):

Summary:

In classical models of signaling network, the signaling activity increases monotonically with the ligand affinity. However, certain receptors prefer ligands of intermediate affinity. In the paper, the authors present a new minimal model to derive generic conditions for ligand specificity. In brief, this requires multi-site phosphorylation and that high-affinity complexes be more prone to degrade. This particular type of kinetic discrimination allows to overcome equilibrium constraints.

Strengths:

The model is simple, and it adds only a few parameters to classical generic models. They moreover vary these additional parameters in ranges based on experimental observations. They explain how the introduction of these new parameters is essential to ligand specificity. Their model quantitatively reproduces the ligand specificity of a certain receptor. They finally provide testable prediction.

Weaknesses:

The naming of multiple variables as activity without precise definitions may be confusing to readers.

Comments on revisions:

I thank the authors for addressing my comments. One point remains regarding the naming of multiple variables as activity. Besides using other words, the authors may consider giving precise definitions of terms, e.g. by writing "We define kinase activity as the phosphorylation rate $\omega = k_p \tau$." A connection that appears only at line 204 in the present manuscript.

<https://doi.org/10.7554/eLife.107524.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

The authors study the steady-state solutions of ODE models for molecular signaling involving ligand binding coupled to multi-site phosphorylation at saturating ligand concentrations. Although the results are in principle general, the work highlights the receptor tyrosine kinases (RTK) as model systems. After presenting previous ODE model solutions, the authors present their own "kinetic sorting" model, which is distinguished by ligand-induced phosphorylation-dependent receptor degradation and the property that every phosphorylation state is signaling competent. The authors show that this model recovers the two types of non-monotonicity experimentally reported for RTKs: maximum activity for intermediate ligand affinity and maximum activity for intermediate kinase activity.

The main contribution of the work is in demonstrating that their model can capture both types of non-monotonicity, whereas previous models could at most capture non-monotonicity of ligand binding.

Strengths:

The question of how energy-dissipating, and thus non-equilibrium, molecular systems can achieve steady-state solutions not accessible to equilibrium systems is of fundamental importance in biomolecular information processing and self-organization. Although the authors do not address the energy requirements of their non-equilibrium model, their comparative analysis of different alternative non-equilibrium models provides insight into the design choices necessary to achieve non-monotonic control, a property that is inaccessible at equilibrium.

The paper is succinctly written and easy to follow, and the authors achieve their aims by providing convincing numerical solutions demonstrating non-monotonicity over the range of parameter values encompassing the biologically relevant regime.

Weaknesses:

(1) A key motivating framework for this work is the argument that the ability to tune to recognize intermediate ligand affinities provides a control knob for signal selection that is available to nonequilibrium systems. As such, this seems like a compelling type of ligand selectivity, which is a question of broad interest. However, as the authors note in the results, the previously published "limited signaling model" already achieves such non-monotonicity in ligand binding affinity. The introduction and abstract do not clearly delineate the new contributions of the model.

We thank the reviewer for this comment. We apologize for any unclear language on our part. The purpose of our work was not to identify the unique reaction scheme to obtain nonmonotonic dependence of network activity on ligand affinity and kinase activity. Rather, we were interested in exploring how such a dependence could arise from the interplay between two ubiquitous network motifs (multisite phosphorylation and active receptor degradation). Notably, as the reviewer later points out, previous models that incorporate only multisite phosphorylation only capture the non-monotonic dependence of network activity on ligand affinity and not kinase/phosphatase activity. We have now clarified this in the abstract (lines 14-16) and the introduction (lines 55-59).

The novel benefit of the model introduced by the authors is that it also achieves a nonmonotonic response to kinase activity. Because such non-monotonicity is observed for RTK, this would make the authors' model a better fit for capturing RTK behavior. However, the broad significance of achieving non-monotonicity to kinase activity is not motivated or supported by empirical evidence in the paper. As such, the conceptual significance of the modified model presented by the authors is not clear.

We thank the reviewer for this comment. We agree that the ability of our model to reproduce non-monotonic dependence on kinase/phosphatase activity was not sufficiently motivated in the original submission. We have now added a brief mention of the biological motivation for nonmonotonic kinase activity in the discussion (lines 229-247) to describe the potential biological significance of this behavior. In particular, non-monotonic kinase/phosphatase dependence may act as a safeguard, filtering out signaling cells with abnormally elevated kinase activity or suppressed phosphatase activity. In the presence of non-monotonic dependence on network activity, downstream signaling would remain contingent on extracellular cues, and cells with extreme kinase/phosphatase imbalances would fail to signal. This could prevent persistent, cueindependent activation, an especially important protective mechanism in pathways regulating metabolically taxing functions such as growth, proliferation, or mounting immune responses. Although direct experimental evidence for the widespread use of this mechanism is currently scarce, our motivation is supported both by the presence of similar regulatory behaviors of phosphatases which arise through distinct mechanisms (such as CD45 in T-cell receptor signaling, (Weiss, 2019)), but highlight the potential biological use of this strategy and by theoretical work on phosphorylation-dephosphorylation cycles, which demonstrates a similar effect in more general settings (Swain, 2013).

(2) Whereas previous models used in the literature are schematized in Figure 1, the model proposed by the authors is missing (see line 97 of page 3). Without the schematic, the text description of the model is incomplete.

We thank the reviewer for identifying this oversight, it has been corrected. See Figure 3 in the new text.

(3) The authors use the activity of the first phosphorylation site as the default measure of activity. This choice needs to be justified. Why not use the sum of the activities at all sites?

We thank the reviewer for this comment. We in fact study all sites (Figure 5A in the resubmitted manuscript). Notably, as suggested by the reviewer, the concentration of the first site is indeed represented by the sum of concentrations of all phosphorylated species. The concentration of the 2nd site is represented by the sum of concentrations of all species except for the first one and so on (lines 153-155).

Reviewer #2 (Public review):

Summary:

In classical models of signaling networks, the signaling activity increases monotonically with the ligand affinity. However, certain receptors prefer ligands of intermediate affinity. In the paper, the authors present a new minimal model to derive generic conditions for ligand specificity. In brief, this requires multi-site phosphorylation and that high-affinity complexes be more prone to degrade. This particular type of kinetic discrimination allows for overcoming equilibrium constraints.

Strengths:

The model is simple, and it adds only a few parameters to classical generic models. Moreover, the authors vary these additional parameters in ranges based on experimental observations. They explain how the introduction of these new parameters is essential to ligand specificity. Their model quantitatively reproduces the ligand specificity of a certain receptor. Finally, they provide a testable prediction.

Weaknesses:

The naming of certain variables may be confusing to readers.

We apologize for the confusion due to unclear presentation. We have clarified our definitions throughout the manuscript.

Reviewer #1 (Recommendations for the authors):

(1) The abstract and introduction present the problem as if this model is solving the fundamental problem of non-monotonic dependence on ligand affinity. However, as the authors noted in their results, this problem has already been solved by a previous phosphorylation model with N-state degradation. What the authors' new model achieves is the additional experimentally observed non-monotonicity of kinase activity dependence. The abstract and introduction should be changed to reflect the actual novel contributions and also to motivate the biological significance of non-monotonic kinase activity dependence.

We thank the reviewer for this comment. We apologize for any unclear language on our part. The purpose of our work was not to identify the unique reaction scheme to obtain nonmonotonic dependence of network activity on ligand affinity and kinase activity. Rather, we were interested in exploring how such a dependence could arise from two ubiquitous network motifs (multisite phosphorylation and active receptor degradation). Notably, as the reviewer later points out, previous models that incorporate only multisite phosphorylation only capture the nonmonotonic dependence of network activity on ligand affinity and not kinase/phosphatase activity. We have now clarified this in the abstract (lines 14-16) and the introduction (lines 55-59). We have also provided biological motivation behind nonmonotonic kinase activity dependence (lines 229-247).

(2) It is important to show (in the supplemental materials if needed) that the closest equilibrium analog to the model (for example, reversible rate constants from each of the activated states to an inactive state) does not achieve non-monotonicity with ligand affinity.

We have added a model in the supplementary materials that represents a detailed balance Markov chain. In the model, we imagine that ligand bound receptors undergo a series of equilibrium transitions, all characterized by the same activation and inactivation rate. We

show that at saturating ligand levels, the signaling output only depends on the ratio of the activation to the inactivation rate (i.e., the thermodynamic stability of the active site) (lines 466-488).

(3) Schematics for earlier models are described in Figure 1. However, no schematic for the actual model proposed by the authors is shown. This should be added as a subpanel to Figure 1.

We thank the reviewer for identifying our omission of our model schematic. We have included our model schematic as its own figure (Figure 3).

(4) Minor: Figure 1 is referred to as Figure?? In line 97 of page 3.

We thank the reviewer for identifying this error, it has been corrected.

Reviewer #2 (Recommendations for the authors):

(1) There is an inconsistency between Figure 2(a) and Equation (1), it suggests that p_N is $\omega^N/(\omega+\delta)^N$. This makes more sense with the model defined in the supplementary material.

We thank the reviewer for identifying this error. Equation (1) has been updated to reflect the correct relationship.

(2) The figure presenting the model of the authors appears to be missing.

We thank the reviewer for identifying this error, it has been corrected (Figure 3 in the new manuscript).

(3) The authors describe phosphorylation as irreversible in the intro, but then consider reversible phosphorylation in their model, which may be confusing to readers.

We thank the reviewer for identifying this source of possible confusion. We have clarified that dephosphorylation is taken to be a distinct irreversible reaction, see lines 105 - 112.

(4) The authors reuse similar names, e.g., network activity, kinase activity, signaling activity, activity. This is confusing.

We apologize for the confusion. We note that, within the context of our model, there are important distinctions between signaling activity (the amount of signaling competent receptors) and kinase activity (value corresponding to the phosphorylation rate). We have attempted to use these different terms correctly and are happy to make clarifying corrections if there are any places where a term is misused.

(5) Several parameters are defined only in the captions of the figures, such as β and ρ .

We thank the reviewer for identifying this omission, we have added the definitions of β and ρ to the main text (see line 129).

(6) The sentence at line 137 lacks some words: "Below, we kinetic...".

We thank the reviewer for identifying this error, we have added the missing words ("Below, we show how kinetic...").

| (7) *The sentence at line 183 lacks some words: "When kinase activity..."*.

We thank the reviewer for identifying this error. We have now corrected it.

| (8) *Figure 5 is very small.*

We will work with the production team to increase the size of this figure.

<https://doi.org/10.7554/eLife.107524.2.sa0>